Emna Ben Mahmoud

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Profile

Software engineering student specializing in backend architecture and full-stack web development. Currently building foundations in generative AI and LLM-based applications, and eager to contribute to a challenging internship where I can grow technically while delivering real value.

Education

Higher Institute of Applied Sciences and Technology of Sousse (ISSATSo) Sousse, Tunisia
Sept 2022 – Present

Engineering Degree, Software Engineering (National Diploma)

- 5-year curriculum focused on Computer Science, Systems Architecture, and Software Development Life Cycles.
- Includes a 2-year Integrated Preparatory Cycle in Mathematics, Physics, and Computer Science (2022–2024).

Experience

Software Engineering Intern, Clubedge Jul, Aug 2026

- Built a provider-independent Communication Layer integrating Mattermost (domain models, service layer, adapter pattern), with REST API and WebSocket integration for real-time messaging and user/channel synchronization.
- Extended the integration with a Next.js frontend.
- **Technologies:** TypeScript, Node.js, ExpressJS, Next.js, Mattermost SDK, Docker.

Software Engineering Intern, Zoomy Technology & IT Aug 2025

- Built an internship management dashboard for offers, applications, candidate evaluation, and task assignment.
- **Technologies:** ReactJS.

Software Engineering Intern, Lunar TC Jul 2025

- Developed a dynamic drag-and-drop form builder with reusable components and exportable form code.
- **Technologies:** Angular, Tailwind CSS.

Projects

Scolar Process Management System GitHub Repository

- Backend system for managing student/teacher accounts, schedules, and course materials, with an integrated discussion forum.
- **Technologies:** NestJS, Prisma, PostgreSQL, Websockets, Docker.

RAG AI Agent GitHub Repository

- RAG app that answers questions from PDFs with source citations, using an event-driven pipeline (Inngest).
- **Technologies:** Python, FastAPI, Inngest, LlamaIndex, Qdrant, Google Gemini API

Conversational RAG Chatbot GitHub Repository

- Developed a conversational RAG chatbot using document chunking, vector embeddings, ChromaDB, semantic search, conversational memory, and query contextualization with the Google Gemini API.
- **Technologies:** Python , Google Gemini API , ChromaDB, Sentence Transformers

Certifications

AI Engineer Explorer Certificate Course, Udemy (Oct 2025)
Fundamentals of Deep Learning, NVIDIA (Sep 2025)

Technical Skills

Languages: JavaScript, TypeScript, Python, Java, C++, SQL/PL-SQL
Frameworks: ReactJS, NestJS, Express.js,
AI & Machine Learning: RAG , LLM APIs, Machine Learning fundamentals
Tools & Concepts: REST APIs, WebSocket, PostgreSQL, MySQL, Clean Architecture, Git, GitHub, Docker

Leadership & Activities

Head of Human Resources Department, Microsoft Club ISSATSo
Organization Member of GAICA and Winter school of AI & SE

Languages

Arabic: Native | **French:** Professional Proficiency | **English:** Professional Proficiency